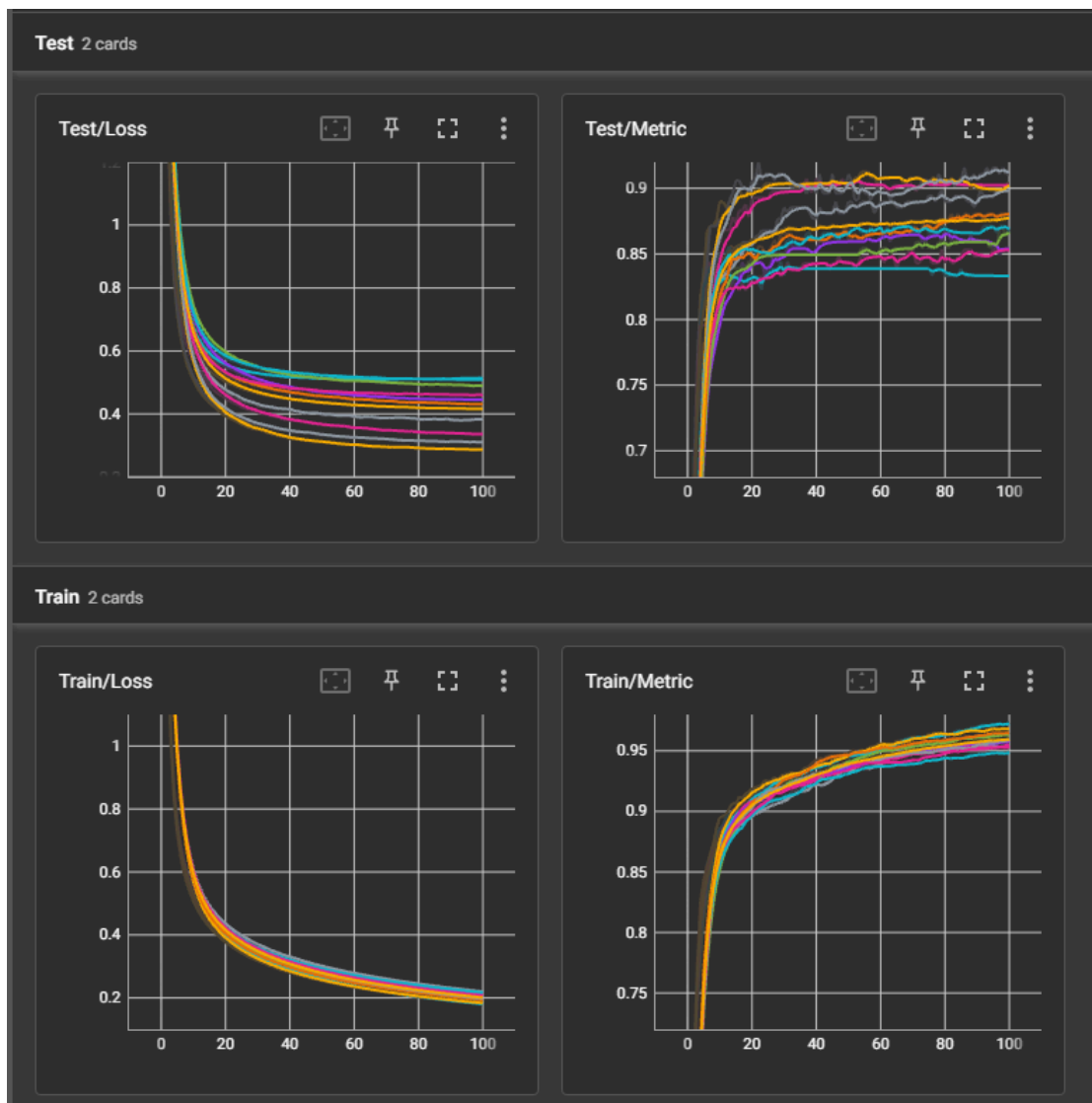


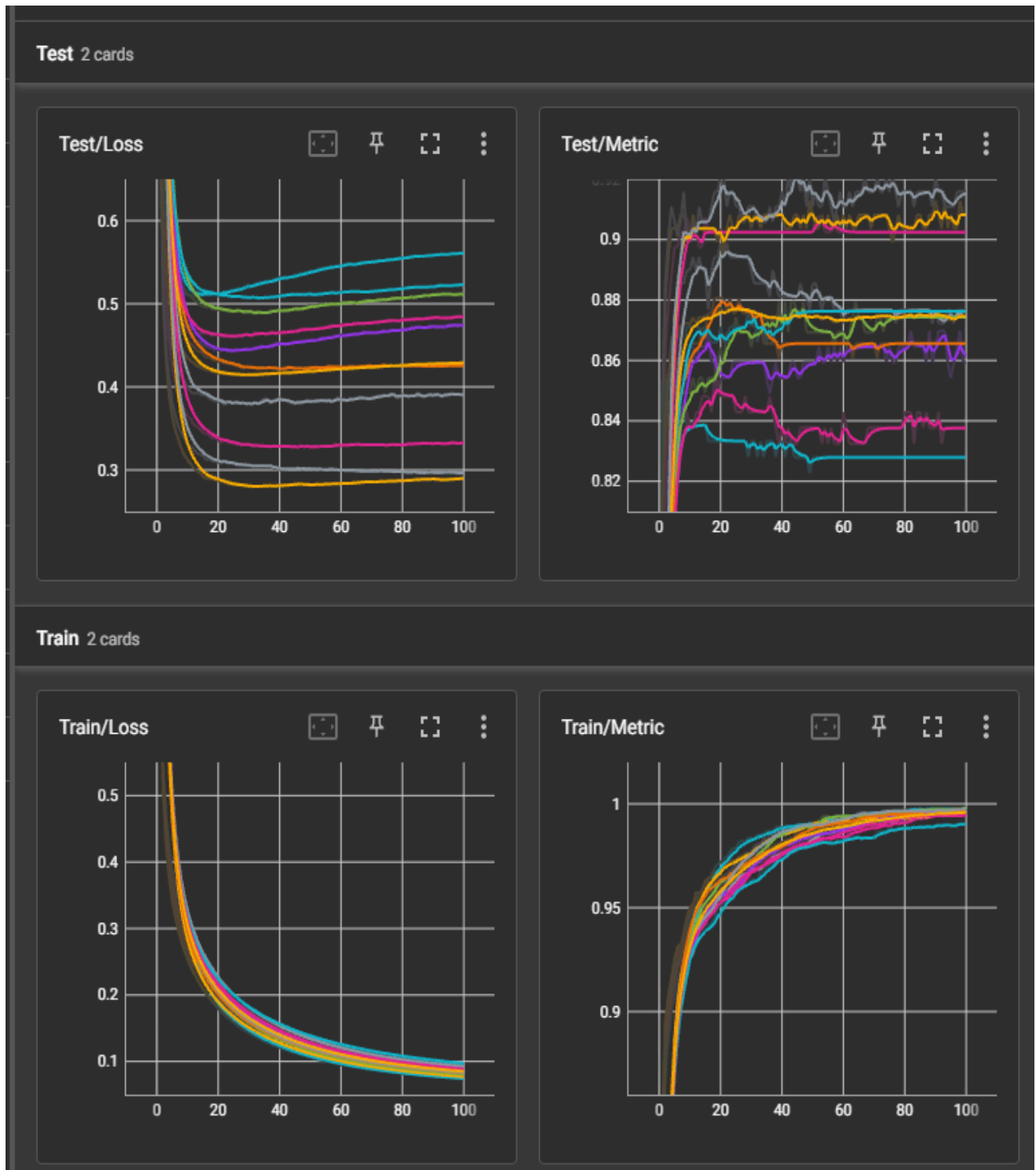
# The Effect of Local Epochs

As seen in the plots below, increasing the number of local epochs results in a clear trend where **training loss decreases rapidly** and **training accuracy reaches near perfection**, while the **test loss and accuracy curves become more unstable** at higher epoch values. For smaller epoch counts (e.g., 1–5), both training and test curves progress smoothly with consistent improvement, indicating stable global updates. However, at larger epoch counts (e.g., 50), the test accuracy curves fluctuate more, and test loss sometimes rises after initial improvement, a sign of **overfitting and loss of global coherence** among client models. These visual results reinforce the conclusion that moderate local epochs yield the most balanced and reliable performance in federated learning.

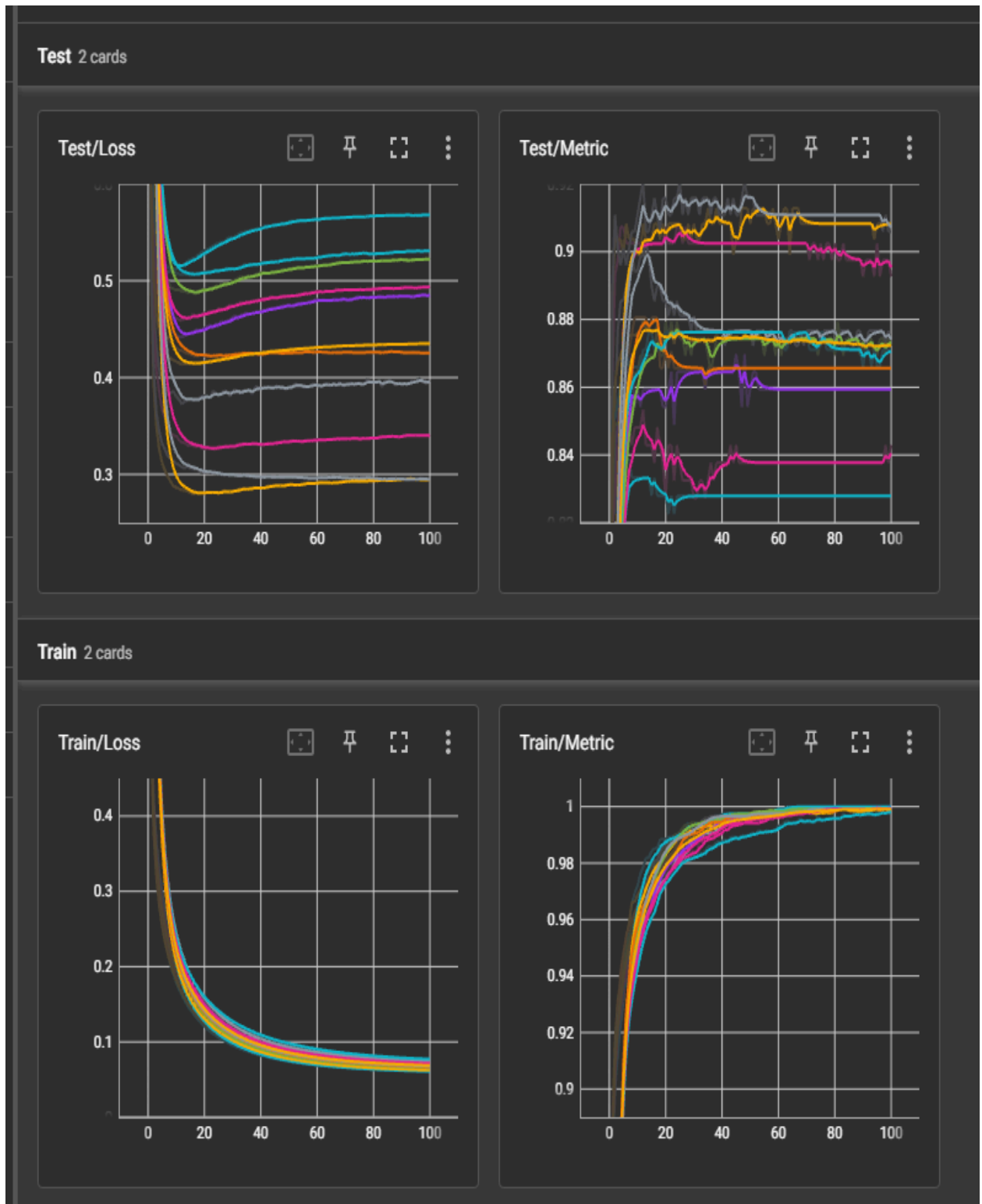
## 1. Local Epochs =1



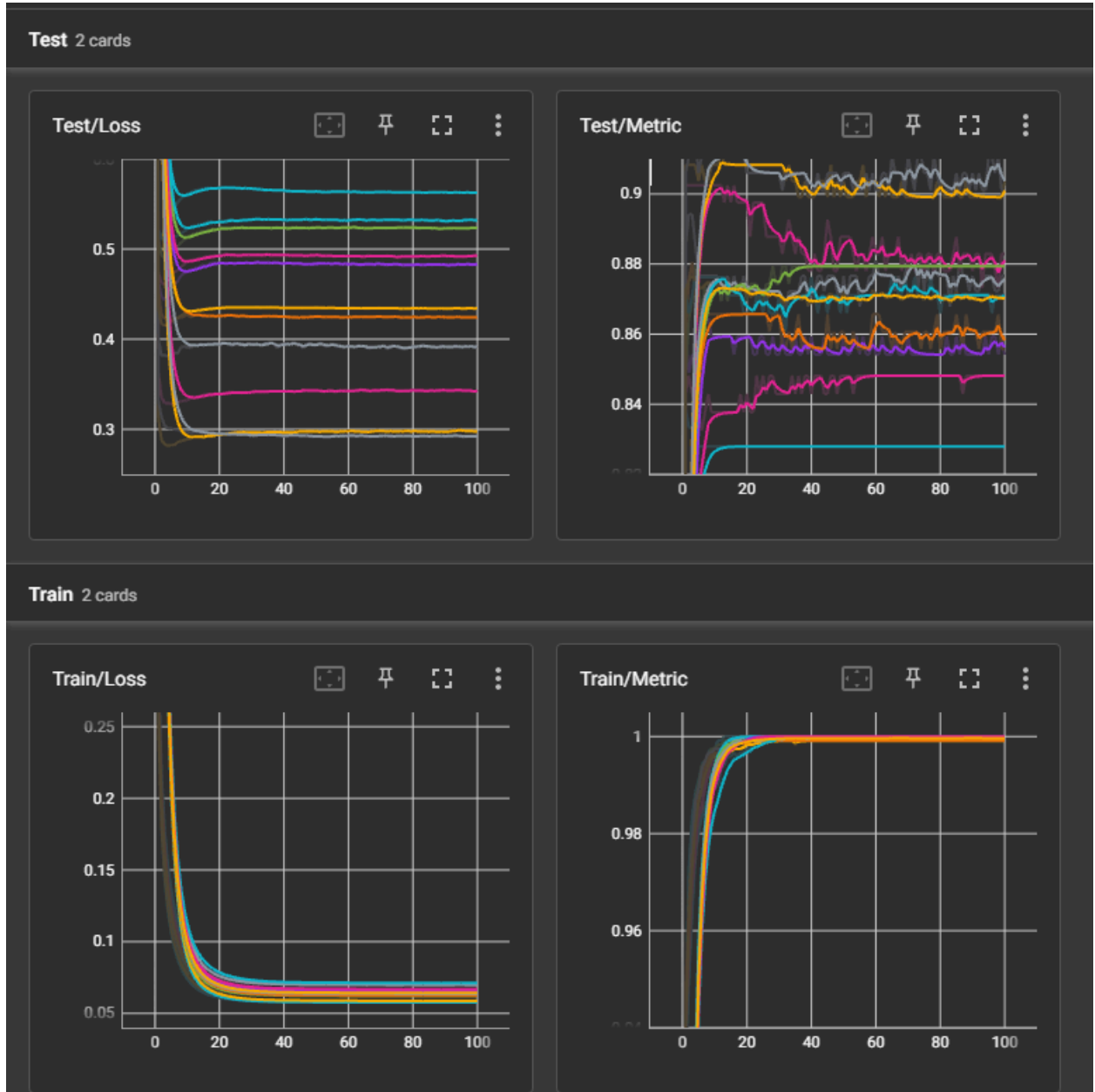
## 2. Local Epochs = 5



### 3. Local Epochs = 10



#### 4. Local Epochs = 50



This result was very much expected based on the theoretical behavior of the **FedAvg** algorithm in federated learning. Increasing the number of local epochs allows each client to train more extensively on its own data, which speeds up local convergence but also causes **client models to drift apart**, especially when data distributions differ across clients. This drift leads to less effective global averaging and poorer generalization, as reflected by the fluctuating and sometimes lower test accuracy at higher local epochs. Conversely, with fewer local epochs, frequent communication keeps client models more aligned, resulting in **slower but more stable**

**and consistent learning.** Therefore, the observed pattern of faster convergence but weaker generalization with higher local epochs aligns perfectly with the expected trade-off between **communication efficiency** and **model consistency** in federated learning.